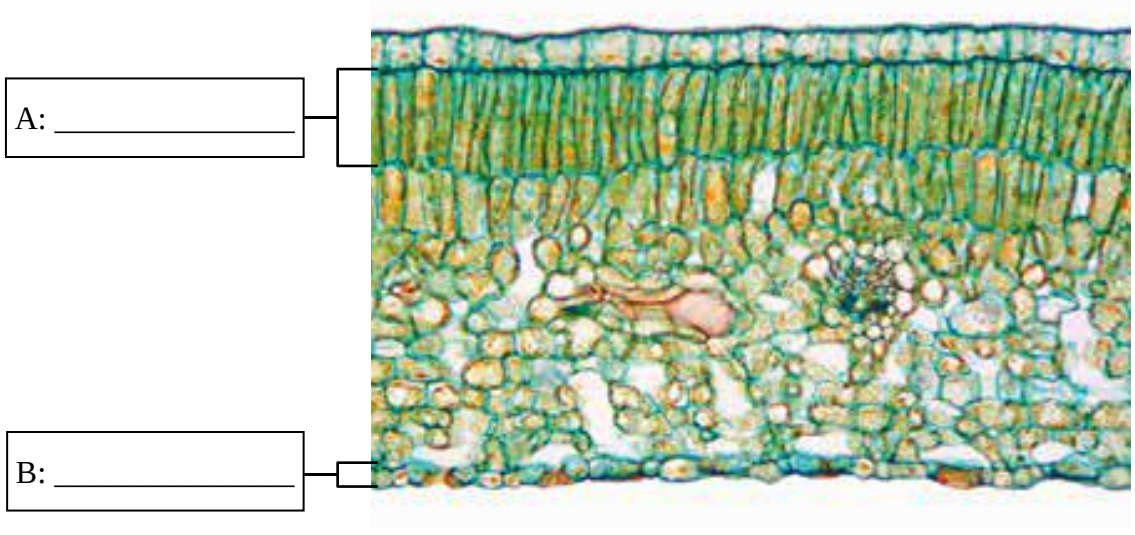
Structured questions

|!|BCELA0210000001|!| ★★

The photomicrograph below shows the transverse section of a leaf of a dicotyledonous plant.



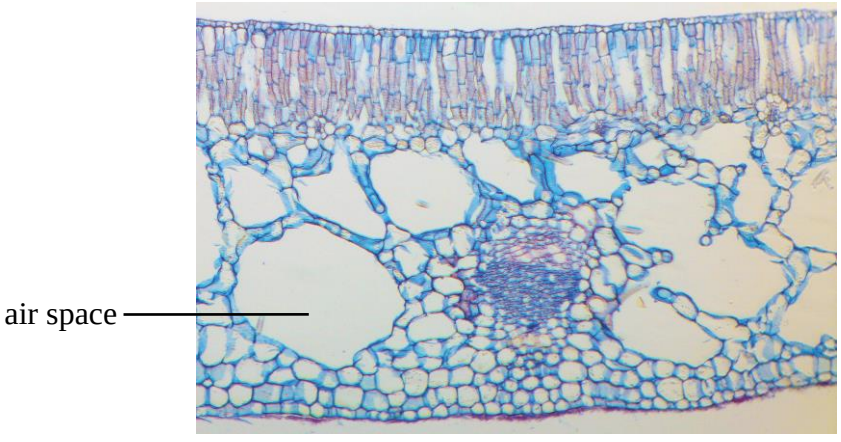
(×100)

(a) Label tissues A and B in the box provided. (2 marks)

(b) Describe ***three*** ways by which this leaf is adapted for light absorption. (3 marks)

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(c) The photomicrograph below is the transverse section of a leaf from a different type of plant.



(×100)

Deduce the habitat of this plant and the importance of having many air spaces.

(4 marks)

##

(a) A: Palisade mesophyll (1)

B: Lower epidermis (1)

(b) Any ***three*** of the following: (max. 3)

- The leaf is broad and flat. This provides a large surface area for absorbing light. (1)
- The leaf is thin so that light can reach the photosynthetic cells easily. (1)
- The leaf blade is supported by the midrib and veins so that it spreads out to maximize the surface area exposed to sunlight. (1)
- The cuticle and the epidermis are transparent so that light can penetrate and reach the photosynthetic mesophyll beneath them. (1)
- The palisade mesophyll is located in the upper part of a leaf. These cells are closely packed and contain numerous chloroplasts. These help maximize the amount of light absorbed for photosynthesis. (1)

(c) This plant is likely to live in a pond / stream / or other aquatic habitat. (1)

The presence of many air spaces helps the leaf to float (1)

because air is less dense than water. (1)

This allows the leaf to absorb more light for photosynthesis. (1)

(9 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	3
(c)	Applying concepts	4

|!|BCELA0210000002|!| ★★ **SI**

Chloroplast suspension and DCPIP solution can be used to investigate the photochemical reactions of photosynthesis. DCPIP is a blue dye. When it gains electrons or hydrogen, it is reduced and turns colourless.

DCPIP (blue) $\xrightarrow{\text{chloroplast suspension in light}}$ reduced DCPIP (colourless)

A student set up three test tubes in different conditions:

- Tube A: 1 cm^3 of buffer solution and 9 cm^3 of DCPIP solution illuminated in light
- Tube B: 1 cm^3 of chloroplast suspension and 9 cm^3 of DCPIP solution in darkness
- Tube C: 1 cm^3 of chloroplast suspension and 9 cm^3 of DCPIP solution illuminated in light

The three test tubes were left at 25 ^oC for 20 minutes. The colour of the mixtures at start and after 20 minutes is recorded as follows:

Tube	Colour of the mixture	
	at start	after 20 minutes
A	greenish blue	greenish blue
B	greenish blue	greenish blue
C	greenish blue	green

(a) The solution used to prepare the chloroplast suspension had the same water potential as the chloroplast. Explain the significance for this. (2 marks)

(b) How are hydrogen atoms generated in the photochemical reactions of photosynthesis? (2 marks)

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(c) Explain for the results of tube C. (3 marks)

(d) Explain why tubes A and B were necessary as controls. (3 marks)

##

(a) To ensure there will not be net water gain or loss from the chloroplasts in suspension. (1)
Otherwise, the chloroplasts may burst or shrink. (1)

(b) Some of the light energy absorbed by chlorophyll is used to split (1)
water molecules into hydrogen and oxygen. (1)

(c) Hydrogen / electrons released in the photochemical reactions (1)
reduce DCPIP to become colourless. (1)
The green colour resulted is due to the chloroplasts present. (1)

(d) Tube A can be used to show that light alone does not cause the colour change in DCPIP (1)
and chloroplasts are required for photochemical reactions to occur. (1)
Tube B can be used to show that light is required for photochemical reactions to occur. (1)

(10 marks)

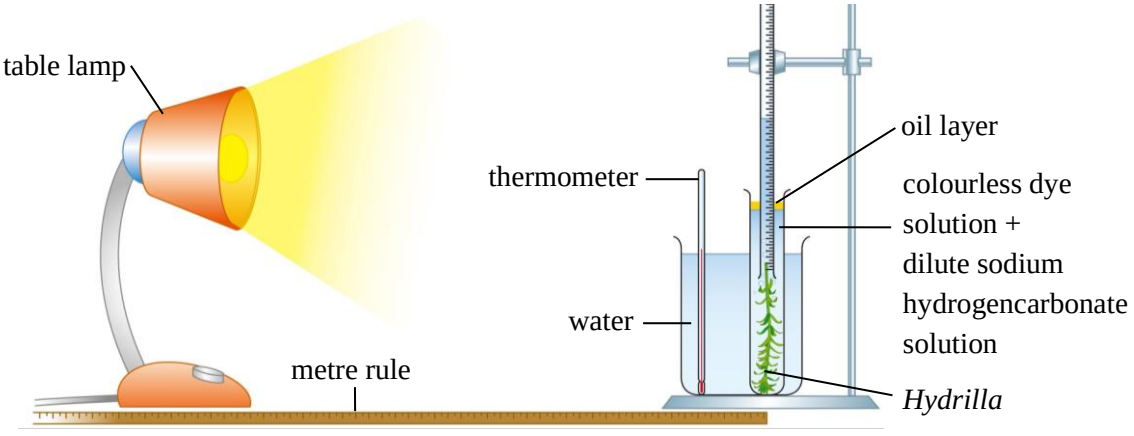
##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Understanding basic concepts	2
(c)	Analysing data	1
	Applying concepts	2
(d)	Applying concepts	3

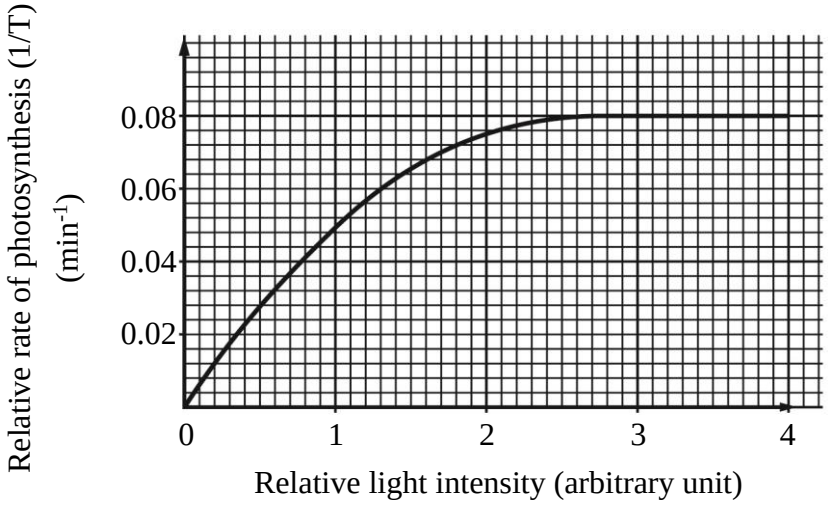
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|!|BCELA0210000003|!| ★★ **SI**

In an investigation, the effect of light intensity on the rate of photosynthesis of *Hydrilla* was studied using the set-up below. The colourless dye solution changes to blue when oxygen is present. The time taken for the colourless dye solution to change to blue reflects the relative rate of photosynthesis.



The graph below shows the results of the investigation.



(a) State the purpose of

(i) adding a layer of oil on top of the solution. (1 mark)

_____

_____

(ii) adding sodium hydrogencarbonate into the test tube. (1 mark)

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(b) The amount of light emitted from the lamp is fixed. Suggest how the light intensity can be varied during the experiment. (2 marks)

(c) What conclusion can be drawn based on the graph? (2 marks)

(d) How can you test scientifically that the colour change of the dye solution is not caused by the presence of carbon dioxide? You may use the following materials and apparatus:

conical flask, two-hole stopper, glass tubing, rubber tubing,
soda lime, lime water, potassium hydroxide solution

Draw the set-up in the space provided. (4 marks)

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##

(a) (i) To prevent oxygen in the air from dissolving in the solution. (1)

(ii) To provide carbon dioxide to the plant. (1)

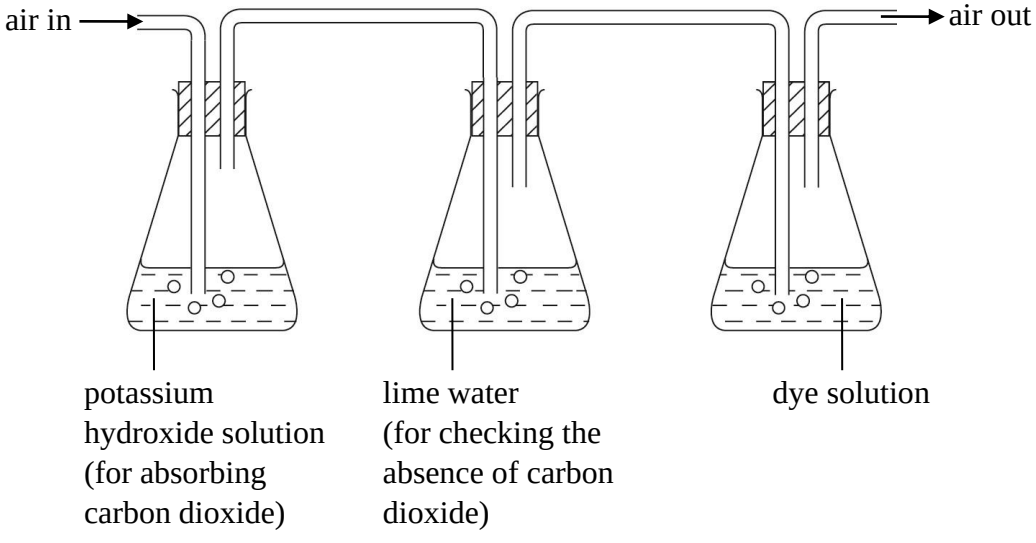
(b) Place the lamp at different distances from the *Hydrilla*. (1)

The closer the lamp to the *Hydrilla*, the higher the light intensity. (1)

(c) The relative rate of photosynthesis increases with increasing light intensity from 0 to 25 arbitrary units. (1)

Above 25 arbitrary units, the rate levels off. (1)

(d)



Soda lime / potassium hydroxide solution in an enclosed container to absorb carbon dioxide (1)

Lime water in an enclosed container to check the absence of carbon dioxide (1)

Dye solution in a container with an air outlet (1)

Correct connection of glass tubing (1)

(10 marks)

##

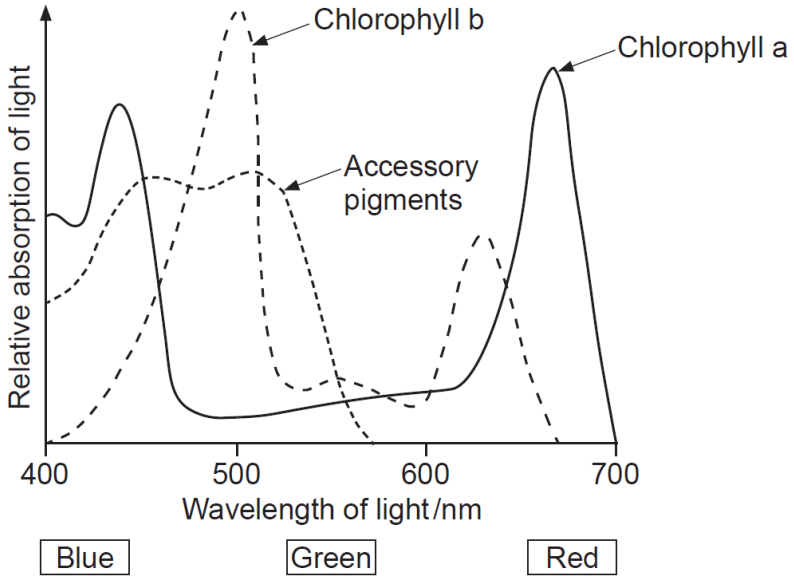
	Skills assessed	Marks
(a)(i)	Applying concepts	1
(a)(ii)	Applying concepts	1
(b)	Applying concepts	2
(c)	Analysing data	2
(d)	Practical skills	4

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|!|BCELB0210000004|!| ★

CCEA GCE AL Biology Assessment Unit A2 2 Jun 2018 Q4

(a) The graph below shows the absorption spectrum for the photosynthetic pigments in a typical plant.

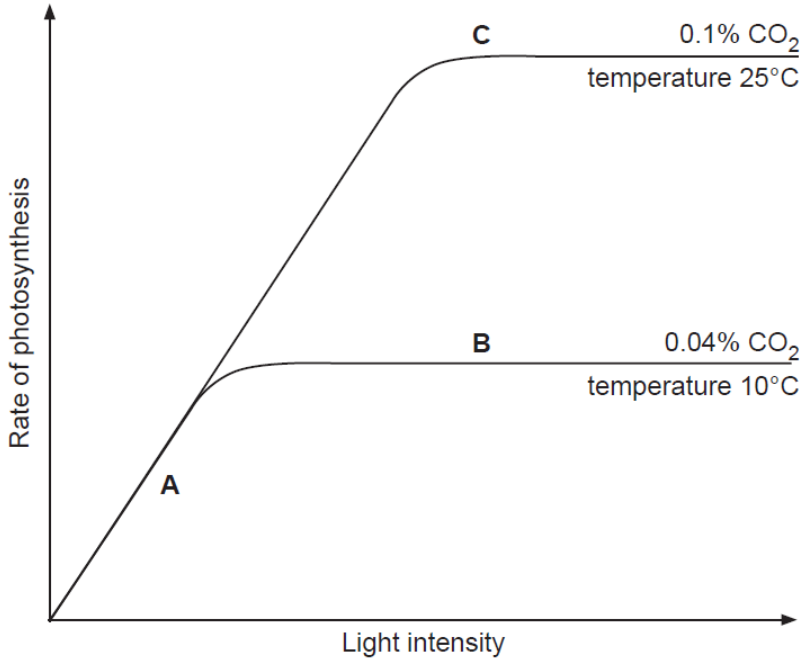


(i) Using the information provided, explain why plant leaves are normally green in colour. (2 marks)

(ii) An action spectrum may also be plotted. Describe what is meant by the term 'action spectrum'. (1 mark)

[This sub-question is out of the HKDSE Biology syllabus.]

(b) The graph below summarizes the relationship between light intensity and rate of photosynthesis in different conditions of temperature and CO_2 availability.



(i) Suggest how scientists investigating this relationship would have accurately measured the rate of photosynthesis. (2 marks)

(ii) Using your knowledge of the photochemical reactions, explain why the rate of photosynthesis is higher at B than at A. (2 marks)

(iii) Using your knowledge of the dark reactions, explain why the rate of photosynthesis is higher at C than at B. (4 marks)

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(c) Many British plant species grow fastest at a temperature of approximately 25 °C. Temperatures greater than this can cause a very high rate of transpiration. The plants' response to this can result in a reduction of growth rate. With reference to the response of plants to high transpiration rates, explain why growth is faster at 25 °C than at 30 °C. (2 marks)

##

(a) (i) Green light is absorbed the least. (1)
It is reflected from the leaves (to give the green colour). (1)

(ii) An action spectrum shows how effective photosynthesis is at different wavelengths. (1)

(b) (i) Use a water plant (or a named water plant) and an oxygen probe in a closed container (or other appropriate apparatus) (1)
to collect the oxygen produced over time in different environmental conditions. (1)

(ii) At higher light intensity, the rate of photochemical reactions is higher, so more electrons are emitted from the electron transport chain. (1)
As a result, more ATP and NADPH are produced for the subsequent reactions (or carbon fixation / Calvin cycle). (1)

(iii) Higher levels of carbon dioxide increase the rates of carbon fixation. (1)
More triose phosphate are produced. (1)
The dark reactions are controlled by many enzymes. (1)
An increased temperature increases enzyme activity and thereby the rates of the dark reactions. (1)

(c) At 30 °C, stomata are closed (as a consequence of water stress), so less carbon dioxide are taken in. (1)
Less glucose / carbohydrate / amino acid / protein can be synthesized for growth. (1)

(13 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
	Applying concepts	1
(a)(ii)	Understanding basic concepts	1
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2
(b)(iii)	Applying concepts	4

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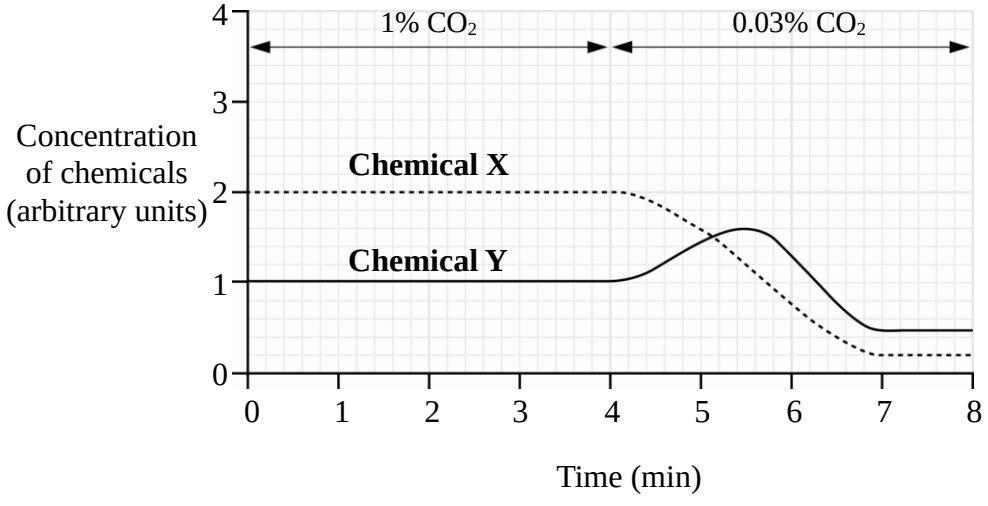
(c)	Applying concepts	2

|!|BCELA0210000005|!| ★★ **SI** **Cross topic**

Paula used a green alga to investigate how changes in carbon dioxide concentrations affect photosynthesis. She illuminated a suspension of green algae with a table lamp and kept the set-up at room temperature. The algal suspension was first supplied with 1% carbon dioxide for four minutes and then 0.03% carbon dioxide for another four minutes.

She recorded the concentrations of chemical X (a 3-C intermediate formed after carbon dioxide is fixed) and chemical Y (a 5-C intermediate) at regular intervals.

The following graph shows the results.



(a) Photosynthesis involves two main stages.
State the main stage in which chemicals X and Y are involved. (1 mark)

_____

(b) (i) State the role of chemicals X and Y in photosynthesis. (2 marks)

_____

_____

_____

(ii) Given the graph above, explain the change in the concentration of chemical Y after four minutes. (2 marks)

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(c) Chemical X can be used in the synthesis of proteins.

(i) State another substance that is required in the synthesis of proteins. (1 mark)

(ii) Explain how the leaf of terrestrial plants obtains the substance given in (c)(i) for use in protein synthesis. (3 marks)

##

(a) Calvin cycle (1)

(b) (i) Chemical X is reduced to triose phosphate (1)

Chemical Y acts as a carbon dioxide acceptor that fixes the atmospheric carbon dioxide (1)

(ii) There is a reduction in the amount of carbon dioxide four minutes into the investigation, so less carbon dioxide is fixed by chemical Y / reacts with chemical Y to form chemical X. (1)

This causes chemical Y to accumulate, thus raising its concentration. (1)

(c) (i) Nitrate / Sulphate (1)

(ii) Nitrate / sulphate ions are absorbed into the root by active transport. (1)

The ions are then transported to the leaf along with water through the xylem. (1)

The flow of water and ions is driven by transpiration pull. (1)

(9 marks)

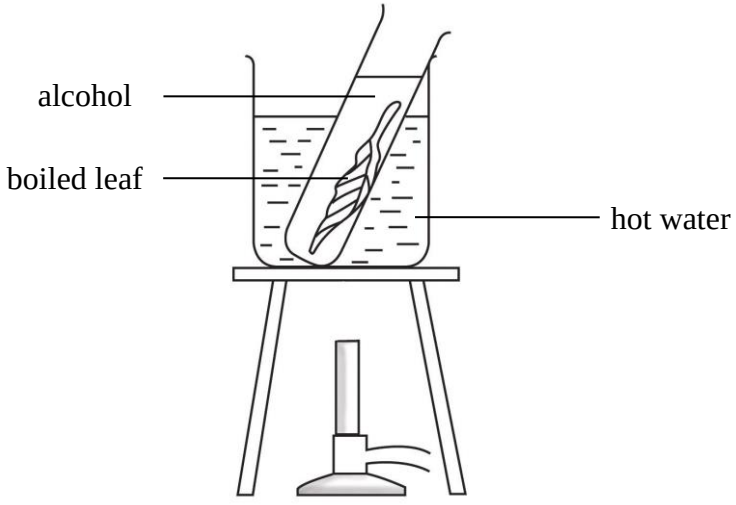
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)(i)	Understanding basic concepts	2
(b)(ii)	Analysing data	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	3

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|!|BCELC0210000006|!| **SI**

The following diagram shows one of the steps in testing a green leaf for starch.



(a) Why should the test tube be heated in a hot water bath instead of being heated directly with a Bunsen flame? (1 mark)

(b) What is the purpose of heating the leaf in alcohol? (1 mark)

(c) Explain why the leaf should be boiled in water before it is heated in alcohol. (2 marks)

(d) What treatment should be given to the leaf after boiling it in alcohol but before adding an iodine solution to it? What is the purpose of this treatment? (2 marks)

(e) What would happen when iodine solution is added to the leaf at the end of this test? (Assume the leaf had been conducting photosynthesis) (1 mark)

##

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(a) The vapour of alcohol would catch fire. (1)

(b) To extract chlorophyll from the leaf / to decolourize the leaf (1)

(c) The leaf was boiled in water in order to

- kill the plant cells, thus stopping their metabolic reactions.
- burst the starch grains.
- make the cells more permeable to alcohol and iodine solution. (Any ***two***, 2)

(d) Put the leaf in hot water again (1)

in order to remove any residual alcohol on the leaf / soften the leaf. (1)

(e) The leaf would turn blue-black. (1)

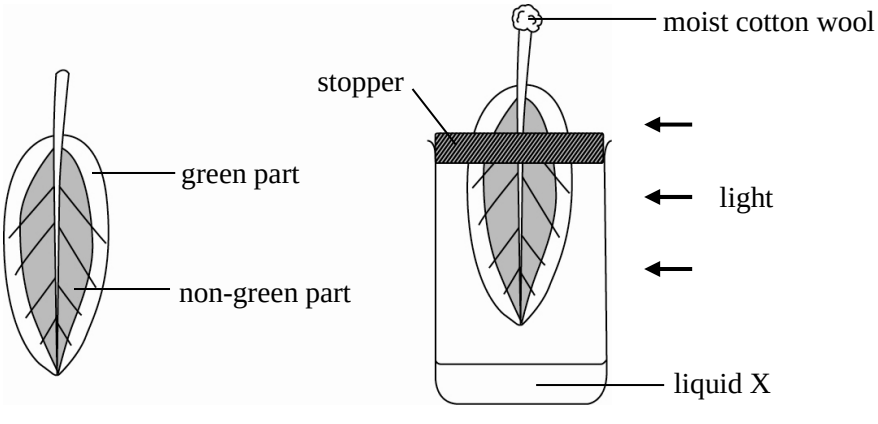
(7 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Practical skills	1
(c)	Practical skills	2
(d)	Practical skills	2
(e)	Understanding basic concepts	1

|!|BCELB0210000007|!| ★ **SI**

An experiment was set up as below in order to investigate the effects of light, carbon dioxide and chlorophyll on photosynthesis. The leaf was detached from a destarched plant.



(a) Name liquid X and state its function. (2 marks)

(b) Draw a diagram to illustrate the colour change in the leaf after an iodine test.

(3 marks)

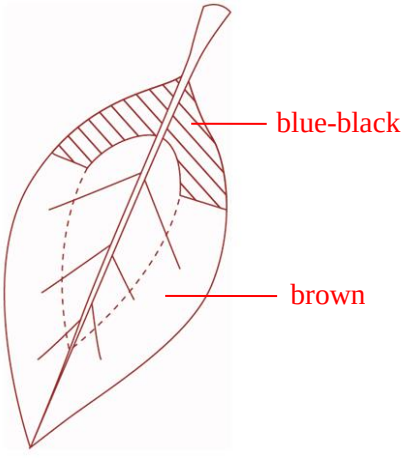
(c) Explain the colour change in the leaf after the iodine test in (b). (5 marks)

##

(a) Liquid X is potassium hydroxide solution. (1)

It absorbs carbon dioxide from the air inside the flask. (1)

(b)



Colour change in a leaf after an iodine test

Correct drawing (1)
Correct labelling (1)
Title (1)

(c) The non-green part (no chlorophyll), (1)
the green part caught by the cork stopper (no incident light) and (1)
the green part inside the flask (no carbon dioxide), (1)
all could not conduct photosynthesis due to the absence of chlorophyll, light and carbon dioxide, respectively. (1)
Photosynthesis could take place in the green part outside the flask, as chlorophyll, light and carbon dioxide were all available. (1)

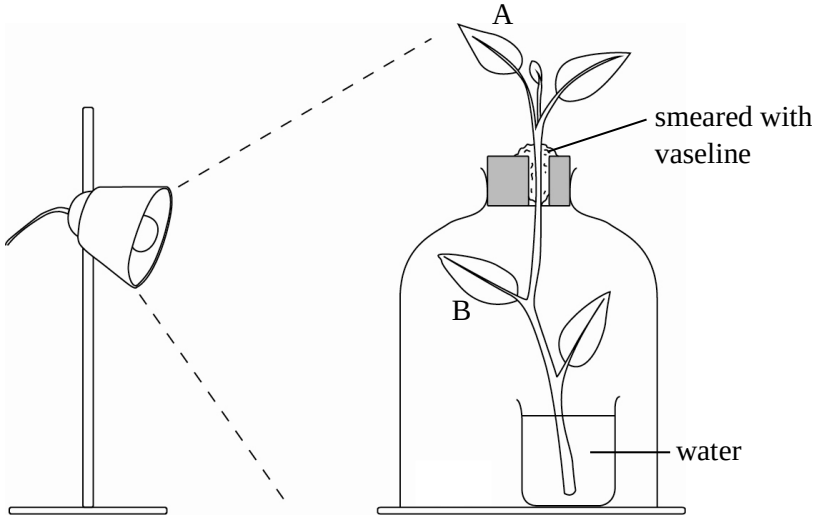
(10 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Presenting information	3
(c)	Applying concepts	5

|!|BCELB0210000008|!| ★ SI

The diagram below shows a set-up used for an experiment to show whether carbon dioxide is necessary for photosynthesis.



(a) What should be done to the plant before setting up the apparatus? Why? (2 marks)

(b) The above apparatus was not set up properly. Suggest and explain ***one*** correction you would make to the set-up. (2 marks)

(c) With the correction stated in (b), which leaf now serves as the control in this experiment? (1 mark)

(d) At the end of the corrected experiment, the leaves will be tested for starch. Name this test and describe what its results will be. (3 marks)

##

(a) The plant should be placed in a dark room for at least 24 hours. (1)
This would destarch the plant / remove the starch from the leaves. (1)

(b) A beaker of soda lime / potassium hydroxide solution should be placed inside the bell jar (1)
in order to absorb the carbon dioxide inside the jar. (1)

(c) Leaf B (1)

(d) Iodine test (1)
Leaf A will turn blue-black (photosynthesis was carried out). (1)
Leaf B will be stained brown (photosynthesis was not carried out). (1)

(8 marks)

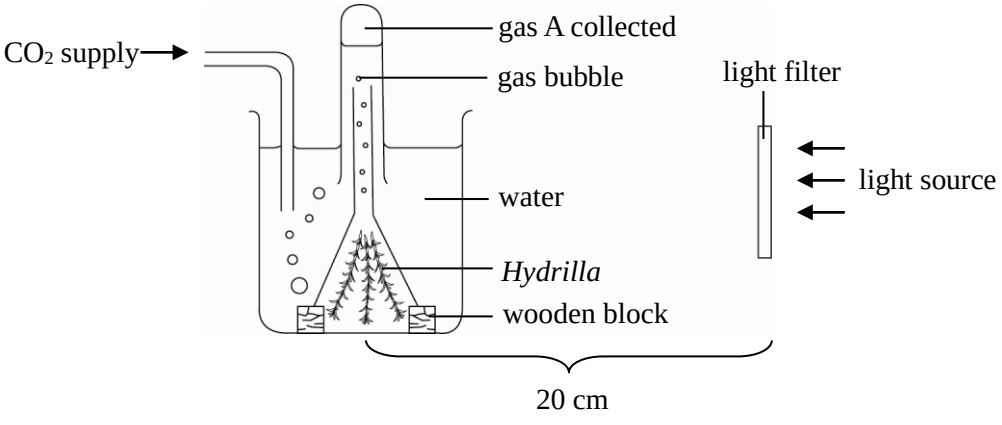
##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Practical skills	2
(c)	Practical skills	1
(d)	Understanding basic concepts	1
	Applying concepts	2

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|!|BCELB0210000009|!| ★ **SI**

The following diagram shows an experimental set-up used to investigate the effect of different light colours on photosynthesis.



A *Hydrilla* plant was covered with an inverted funnel. Bubbles of gas A were released from the cut end of the *Hydrilla* shoot. The number of bubbles evolved per minute under different light filters was recorded. The results are shown below.

Type of light source	**Number of bubbles evolved per minute**
Red light	52
Yellow light	4
Blue light	48
Green light	0

(a) What other method can be used to supply carbon dioxide to the *Hydrilla* shoots apart from pumping it directly into the water? (1 mark)

_____

_____

(b) In the experiment above, an aquatic plant was used instead of a terrestrial plant. Why? (2 marks)

_____

_____

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(c) What is the function of the wooden blocks in this experiment? (2 marks)

(d) Gas A is produced during the experiment.

(i) Name gas A. (1 mark)

(ii) How can you test the identity of gas A? (2 marks)

(e) Referring to the table above, what colour(s) of light would lead to a relatively significant rate of photosynthesis in the *Hydrilla* shoots? (1 mark)

(f) What would happen to plants if only green light reaches the Earth? (2 marks)

##

(a) Adding sodium hydrogencarbonate into the water (1)

(b) An aquatic plant can relatively easily provide a quantitative result (1)
through a discrete number of bubbles evolved per minute. (1)

(c) The wooden blocks raise the funnel and allow a continuous flow of water through. (1)
This ensures that carbon dioxide is constantly supplied to the *Hydrilla* shoots. (1)

(d) (i) Oxygen (1)

(ii) Expose a glowing splint to gas A. (1)
The splint would relight in the presence of oxygen. (1)

(e) Red light and blue light (1)

(f) Photosynthesis would not occur. (1)
Green plants would show poor growth and might die. (1)

(11 marks)

##

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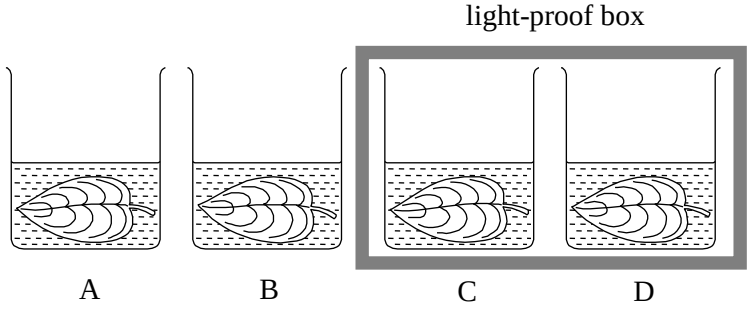
	Skills assessed	Marks
(a)	Practical skills	1
(b)	Practical skills	2
(c)	Practical skills	2
(d)(i)	Understanding basic concepts	1
(d)(ii)	Practical skills	2
(e)	Analysing data	1
(f)	Applying concepts	2

|!|BCELB0210000010|!| ★ **SI**

An investigation was carried out using the leaves from an aquatic dicotyledonous plant. The procedures were as follows:

Procedure 1: The plant was left in a dark room for 48 hours.

Procedure 2: Four leaves were detached from the plant and each of them was placed in the set-ups shown below.



Set-up	Content
A	unboiled distilled water
B	boiled distilled water
C	unboiled distilled water
D	boiled distilled water

Procedure 3: The set-ups were placed under sunlight for four hours.

Procedure 4: The four leaves were tested using the iodine test.

(a) State ***one*** factor that needed to be controlled in this experiment. (1 mark)

_____

(b) One more procedure should have been carried out before procedure 2.

(i) State this procedure. (1 mark)

(ii) State the purpose of this procedure. (2 marks)

(c) Which set-ups can investigate whether the following factors are necessary for photosynthesis?

(i) Light (1 mark)

(ii) Carbon dioxide (1 mark)

(d) State the colour of each leaf after the iodine test. (2 marks)

##

(a) Temperature / leaf size / other reasonable answers (1)

(b) (i) Before procedure 2, detach one leaf from the plant and carry out an iodine test. (1)

(ii) The procedure shows whether the leaves used in the investigation were free of starch. (1)

This makes it possible to determine if any starch detected at the end of the investigation was produced during the experiment. (1)

(c) (i) A and C (1)

(ii) A and B (1)

(d) A: blue black;

B: brown;

C: brown;

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D: brown

($^1/_2$ each, total 2)

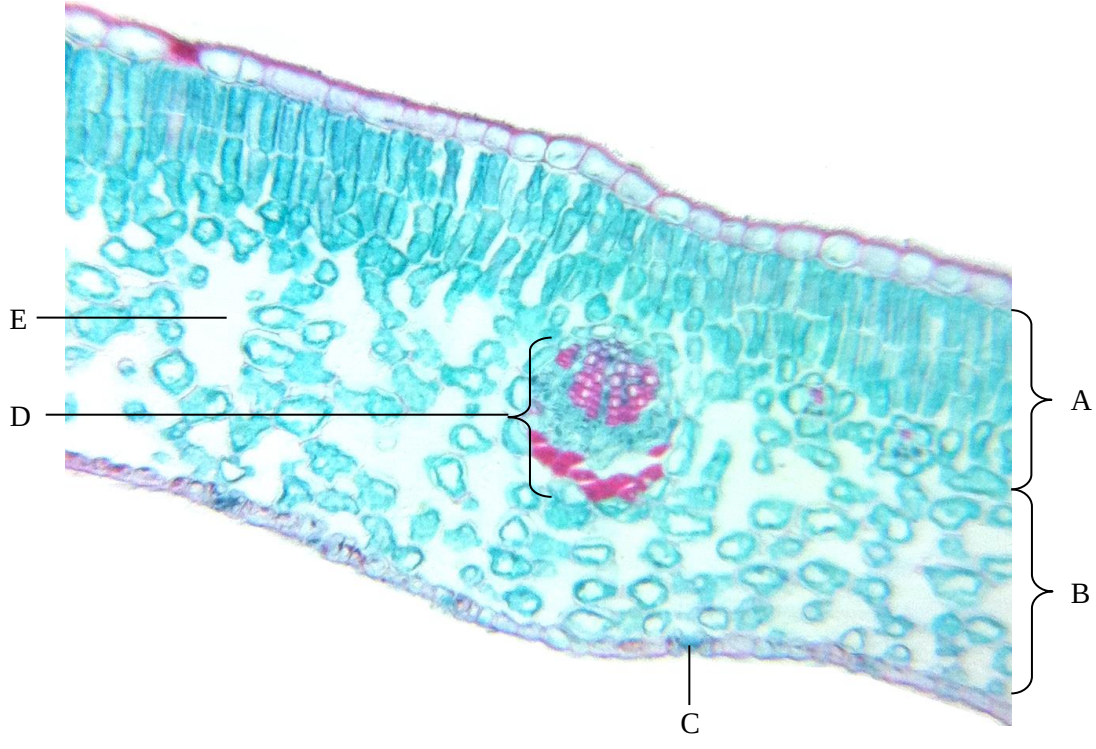
(8 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)(i)	Practical skills	1
(b)(ii)	Practical skills	2
(c)(i)	Analysing data	1
(c)(ii)	Analysing data	1
(d)	Analysing data	2

|!|BCELA0210000011|!| ★★ **Cross topic**

The following photomicrograph shows the transverse section through a dicotyledonous leaf.



(Image source: https://commons.wikimedia.org/wiki/File:Winter_jasmine_leaf_cross_section.png)

(a) State ***two*** structural differences between the cells in layer A and layer B. (2 marks)

(b) The leaf has more of structure C on its lower surface than its upper surface. Describe a simple method other than using a microscope to show this difference. (2 marks)

(c) A layer of lower epidermis is peeled off and observed under a microscope. Draw a diagram with the appropriate title and labels to show structure C and its associated cells as well as features. (4 marks)

(d) Briefly describe and explain the process by which the photosynthetic products made in leaf cells are transported to structure D. (2 marks)

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(e) What is the importance of structure E to photosynthesis? (1 mark)

_____

_____

##

(a) Any ***two*** of the following pairs: (2)

Cells in layer A	***Cells in layer B***
columnar in shape	irregular in shape
closely-packed	loosely packed
have more chloroplasts	have fewer chloroplasts

(b) Put pieces of anhydrous cobalt chloride paper (blue) onto both the upper and lower epidermis. (1)

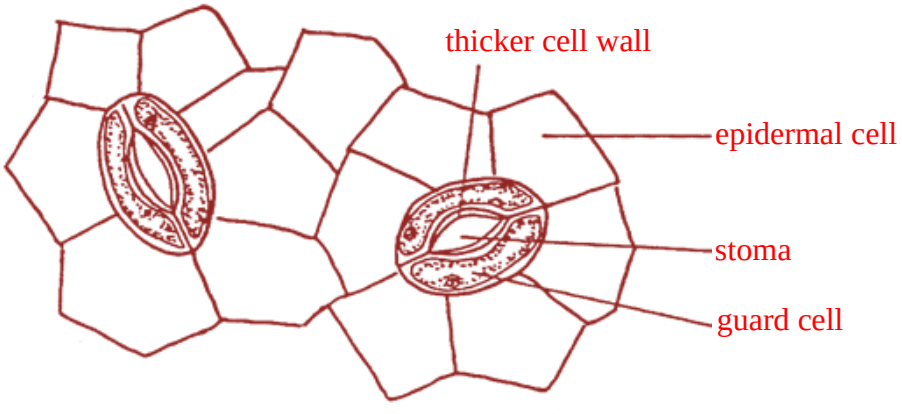
The pieces attached to the lower surface will turn pink first. (1)

OR

Immerse the leaf in a beaker of hot water. (1)

More air bubbles will form on the lower surface of the leaf. (1)

(c)



A diagram showing structure C and the associated cells

Diagram: clear and accurate diagram (1)

Appropriate title (1)

Labels ($^1/_2$ mark each) (2)

(d) The photosynthetic products (sugars) diffuse from the mesophyll cells to the phloem (1)

along the concentration gradient. (1)

(e) Carbon dioxide, which is required for photosynthesis, diffuses through structure C and reaches the mesophyll cells via structure E (air space). (1)

(11 marks)

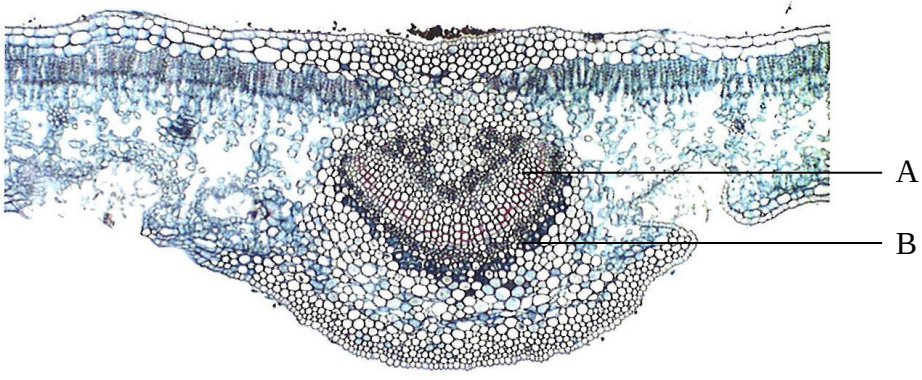
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	Skills assessed	Marks
(a)	Analysing data	2
(b)	Practical skills	2
(c)	Presenting information	4
(d)	Applying concepts	2
(e)	Applying concepts	1

|!|BCELA0210000012|!| ★★ **Cross topic**

The following photomicrograph shows the transverse section of the leaf of a xerophyte.



(a) Describe how the palisade mesophyll is adapted to plant nutrition. (3 marks)

(b) A and B are conducting tissues. Name a substance that is conducted in

(i) tissue A and (1 mark)

(ii) tissue B. (1 mark)

(c) (i) What climate is the xerophyte adapted to? (1 mark)

(ii) State ***one*** feature in the leaf shown above that is an adaptation to this climate. (1 mark)

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##

(a) Palisade mesophyll cells are closely packed and contain abundant chloroplasts. (1)
The palisade mesophyll is located near the upper epidermis and so can receive the maximum amount of light. (1)
Thus photosynthesis can be conducted efficiently to maximize food production. (1)

(b) (i) Water / Minerals (1)

(ii) Sugar / Sucrose / Amino acids (1)

(c) (i) Dry climate (1)

(ii) Any ***one*** of the following: (1)

- Stomata sunken in pits
- Thick waxy cuticle on epidermis

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Analysing data	1

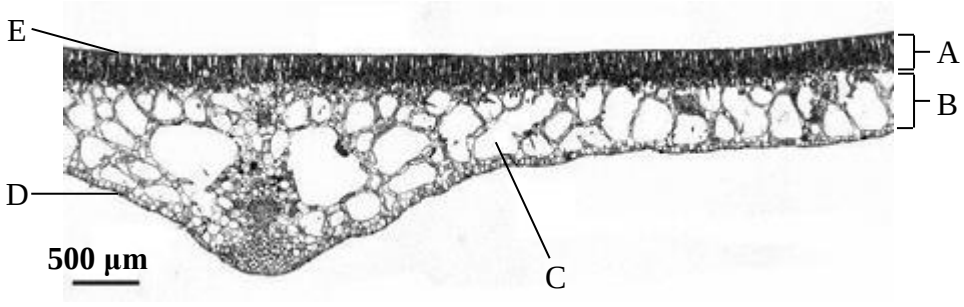
|!|BCELA0210000013|!| ★★ **Cross topic**

Figure I shows the leaves of water lilies and **Figure II** shows the transverse section of one such leaf.

Figure I



Figure II



(a) Referring to **Figure I**, state **two** external features of the leaf that allow it to carry out photosynthesis effectively. Explain briefly. (4 marks)

(b) (i) Compared to a typical dicotyledonous leaf, which of the mesophyll layers in the leaf shown in **Figure II** is thicker than usual? Give your answer using the biological term for this mesophyll layer. (1 mark)

(ii) How is the mesophyll layer stated in the answer to (b)(i) adapted to the water-lily's aquatic environment? (1 mark)

(c) (i) State ***one*** difference between the structures of layers D and E. (1 mark)

(ii) Account for this difference. (2 marks)

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##

(a) The leaf is thin, (1)
which allows gases and light to easily reach the mesophyll cells. (1)
The leaf is broad, (1)
thus providing a large surface area for the absorption of sunlight. (1)

(b) (i) Spongy mesophyll (1)

(ii) The air space between the spongy mesophyll cells holds air within it in order to provide buoyancy, which allows the leaf to float on water. (1)

(c) Any ***one set*** of the following:

(i) The upper epidermis is covered by the cuticle while the lower epidermis is not. (1)

(ii) The cuticle on the upper epidermis prevents excess water loss to the surroundings. (1)
The lower epidermis is submerged in water and does not require a cuticle because there is no danger of desiccation. (1)

OR

(i) The upper epidermis has stomata while the lower epidermis has none. (1)

(ii) The upper epidermis has stomata for gas exchange (1)
while gases dissolved in the water can diffuse into the lower epidermis directly. (1)

(9 marks)

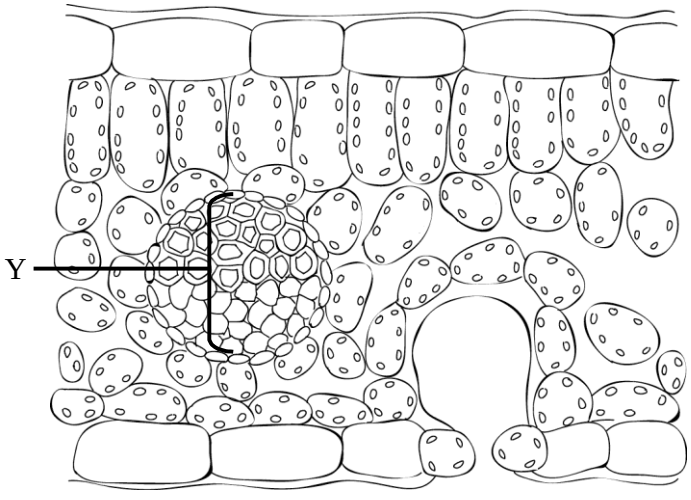
##

	Skills assessed	Marks
(a)	Analysing data	4
(b)(i)	Analysing data	1
(b)(ii)	Applying concepts	1
(c)(i)	Analysing data	1
(c)(ii)	Applying concepts	2

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|!|BCELB0210000014|!| ★

The transverse section of a dicotyledonous leaf is shown below.



(a) With reference to the diagram, state ***two*** structural features of the leaf which usually make the upper leaf surface have a deeper green colour than the lower leaf surface. (2 marks)

(b) Why are the structural features stated in (a) important? (1 mark)

(c) A photosynthetic product is converted to a certain substance and leaves the leaf through a structure in tissue Y. Name the substance and the structure. (2 marks)

(d) Describe and explain what will happen to the substance in (c) when it reaches a fruit. (2 marks)

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(e) Describe how proteins are synthesized from the intermediates of photosynthesis.

(2 marks)

##

(a) The cells in the upper surface of the leaf contain more chloroplasts and (1)
are more closely packed than those on the lower surface. (1)

(b) They allow the maximum amount of light to be absorbed for photosynthesis (1)

(c) The substance: sucrose (1)
The structure: phloem (1)

(d) Sucrose will be converted to starch for storage (1)
as starch is insoluble in water and will not affect the water potential of cells. (1)
(Or other reasonable answers)

(e) The intermediates of photosynthesis combine with the nitrates and sulphates
absorbed from the soil (1)
to form amino acids, which can be used to synthesize proteins. (1)

(9 marks)

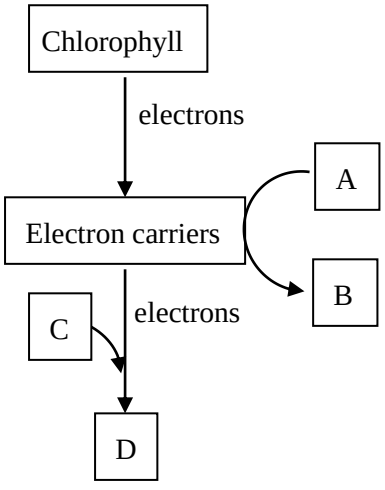
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	1
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2
(e)	Understanding basic concepts	2

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|!|BCELA0210000015|!| ★★

The following diagram illustrates the different stages of the photochemical reactions in photosynthesis.



(a) State the precise location where the photochemical reactions occur in a plant cell. (1 mark)

(b) Identify substances A, B, C and D. (2 marks)

(c) Identify the following processes:

(i) the process that provides electrons to replace those lost by chlorophyll. (1 mark)

(ii) the process that leads to the conversion of A into B. (1 mark)

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(d) Atrazine is a chemical that can be used as a weedkiller. This chemical inhibits the flow of electrons between the electron carriers. Describe and explain how atrazine affects the production of carbohydrates in photosynthesis. (4 marks)

##

(a) Thylakoid membranes (1)

(b) A: ADP
B: ATP
C: NADP
D: NADPH ($^1/_2$ mark each) (2)

(c) (i) Photolysis (1)

(ii) Photophosphorylation (1)

(d) The inhibition of electron flow leads to the reduced production of NADPH. (1)
The generation of ATP is similarly reduced. (1)
Less NADPH and ATP are therefore available to reduce the 3-C compound into triose phosphate in the Calvin cycle. (1)
As a result, fewer carbohydrates are produced. (1)

(9 marks)

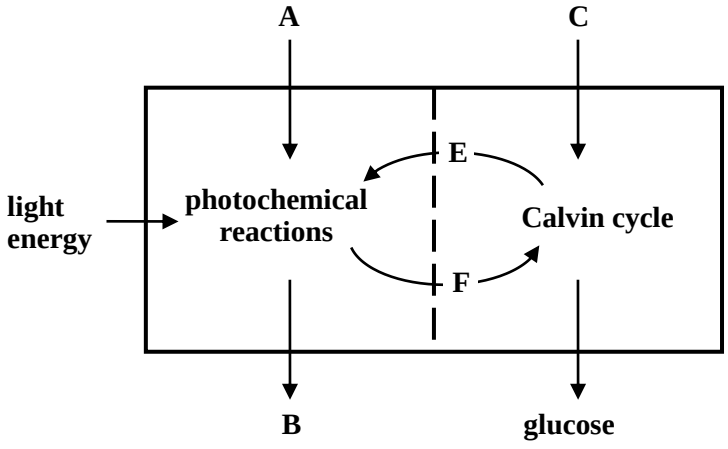
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Analysing data	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1
(d)	Applying concepts	4

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|!|BCELA0210000016|!| ★★

The following diagram summarizes the process of photosynthesis. The rectangle represents the site where photosynthesis occurs in the plant cell.



(a) Name the cell organelle represented by the rectangle. (1 mark)

(b) Describe how substance A is converted to substance B during the photochemical reactions. (3 marks)

(c) Describe a laboratory technique that can be used to show that substance B originates from substance A. (2 marks)

(d) (i) Give ***one*** example of what substances E and F can be, respectively. (2 marks)

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(ii) What is the importance of substance F in the Calvin cycle? (1 mark)

_____

_____

(e) Name the part of the plant through which substance C enters the plant. (1 mark)

_____

##

(a) Chloroplast (1)

(b) Light energy excites some electrons in the chlorophylls to higher energy levels. (1)
As electrons pass from an electron carrier of higher energy level to one of a lower energy level, (1)
energy is gradually released and used to split A (water) into hydrogen and oxygen, the latter of which is released as B (oxygen gas). (1)

(c) When the water (A) supplied to a plant contains ^{18}O isotope, (1)
the isotope will appear in the oxygen (B) released by the plant. (1)

(d) (i) Substance E: ADP / NADP (1)
Substance F: ATP / NADPH (1)

(ii) Any ***one*** of the following: (1)
If ATP was given as F in (d)(i):
provides energy for the reduction of a 3-C compound to triose phosphate
OR
provides energy for the regeneration of a 5-C carbon dioxide acceptor
If NADPH was given as F in (d)(i):
reduces a 3-C compound into triose phosphate

(e) Stoma (1)

(10 marks)

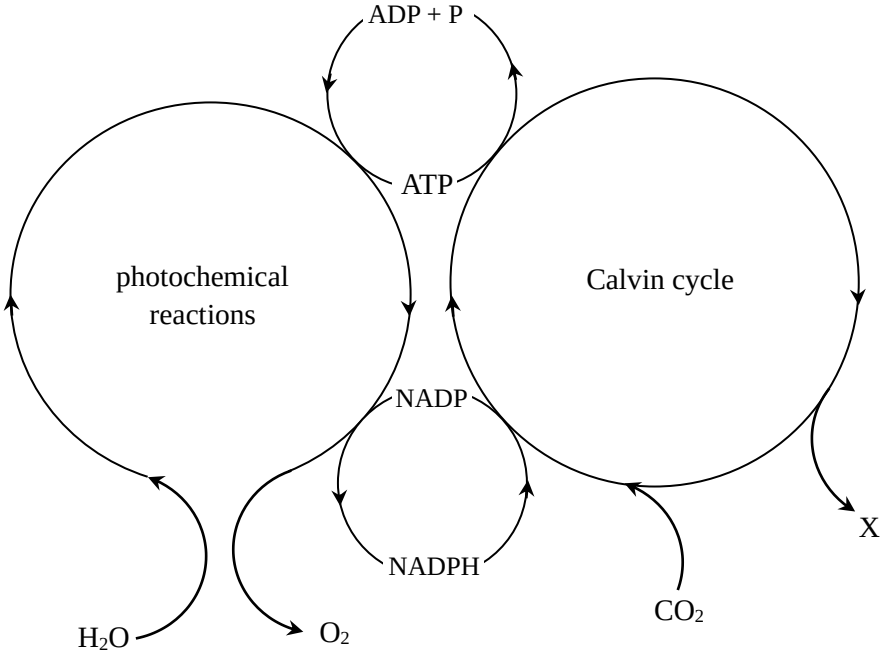
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	3
(c)	Practical skills	2
(d)(i)	Understanding basic concepts	2
(d)(ii)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCELB0210000017|!| ★

During photosynthesis in plants, light energy is converted into chemical energy. The diagram below outlines the processes involved.



(a) Describe how NADPH is produced during photochemical reactions. (4 marks)

(b) During the Calvin cycle, the primary product X is produced.

(i) What is X? (1 mark)

(ii) Explain how carbon dioxide and ATP are used to synthesize X. (3 marks)

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(c) X and certain intermediates produced during photosynthesis can be converted into other important compounds useful to plants. State ***two*** of such compounds and explain how each is useful to plants. (2 marks)

##

(a) Light energy excites certain electrons in chlorophyll molecules to a higher energy level and eventually removes them from the chlorophyll molecules. (1)
Energy is released as these electrons pass down a chain of electron carriers. (1)
Some of the energy is used to split water molecules into oxygen and hydrogen. (1)
The hydrogen is then taken up by NADP to form NADPH. (1)

(b) (i) X is triose phosphate. (1)

(ii) A 5-carbon (5-C) carbon dioxide acceptor combines with carbon dioxide (1)
to form an unstable 6-C compound, which then splits into two molecules of a 3-C compound. (1)
The 3-C compound is then converted into triose phosphate (X) with the use of energy from ATP. (1)

(c) Any ***two*** of the following: (2)

- Glucose for releasing energy
- Cellulose for making cell walls
- Lipids for making cell membranes and energy reserves
- Proteins for growth and repair

(Or other reasonable answers)

(10 marks)

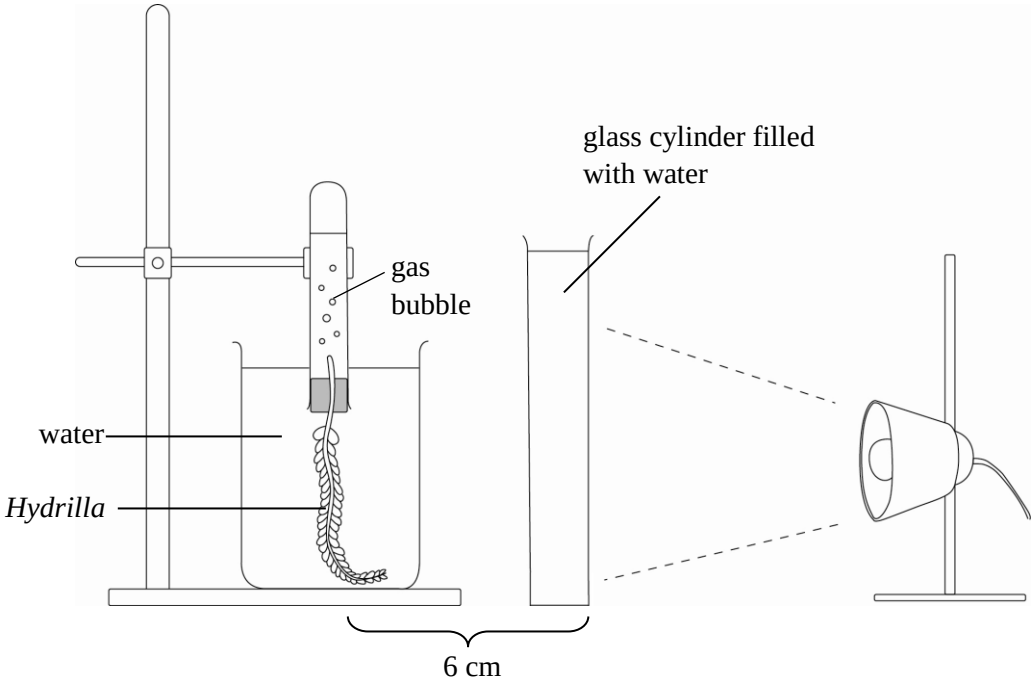
##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	3
(c)	Understanding basic concepts	2

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|!|BCELB0210000018|!| ★ **SI**

The following diagram illustrates an experiment to investigate the effect of light intensity on the rate of photosynthesis in *Hydrilla*.



(a) Suggest a reason for putting the water-filled glass cylinder between the lamp and the *Hydrilla* plant. (1 mark)

_____

_____

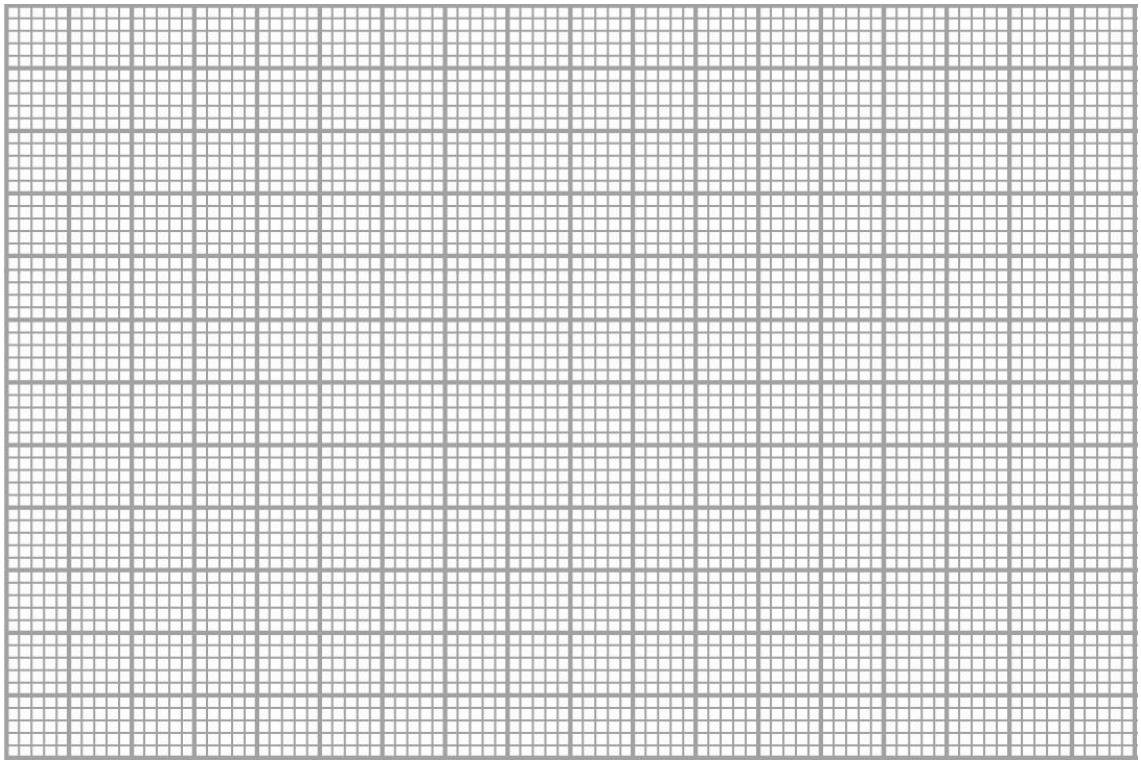
The following table shows the results of this experiment.

Distance of lamp from *Hydrilla* (cm)	**No. of gas bubbles produced per minute by *Hydrilla***
5	30
10	30
15	27
20	24
25	20
30	17
40	11
50	6
60	2

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(b) Plot the data on the graph paper below. Include the appropriate labels and title.

(3 marks)



(c) Account for the graph plotted in (b). (4 marks)

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(d) Explain what would happen to the number of gas bubbles produced per minute by the *Hydrilla* plant if

(i) ice cubes were put into the beaker of water. (2 marks)

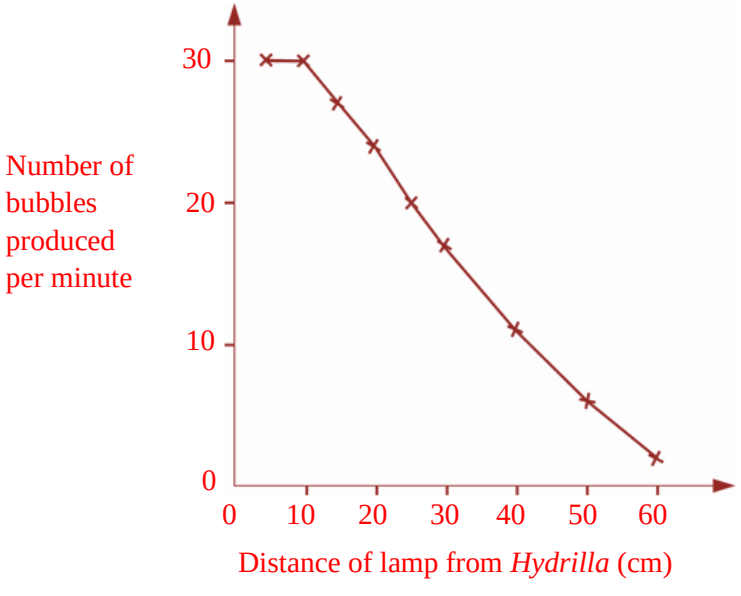
(ii) sodium hydrogencarbonate was added into the beaker of water. (2 marks)

##

(a) The heat released by the lamp is absorbed by the water in the glass cylinder, thereby controlling temperature as a constant factor in the experiment. (1)

(b) Correct title (1)
Correct labelling of axes (1)
Correct plotting and joining of points (1)
e.g.

The effect of light intensity on the rate of photosynthesis



(c) As the distance of the lamp from the *Hydrilla* plant increases from 10 cm to 60 cm, the light intensity decreases. (1)
Less light energy is provided for photosynthesis / light becomes the limiting factor of photosynthesis, and hence the number of gas bubbles produced per minute decreases. (1)
When the distance between the lamp and the *Hydrilla* plant is 5-10 cm, the light intensity is very high and it is not the limiting factor of photosynthesis. (1)

Other factors now limit the rate of photosynthesis, and so the number of gas bubbles produced per minute is almost constant. (1)

(d) (i) The gas bubbles produced per minute would decrease (1)
because the rate of photosynthesis decreases with lower temperatures. (1)

(ii) The gas bubbles produced per minute would increase (1)
because sodium hydrogencarbonate provides additional carbon dioxide, which is required for photosynthesis. (1)

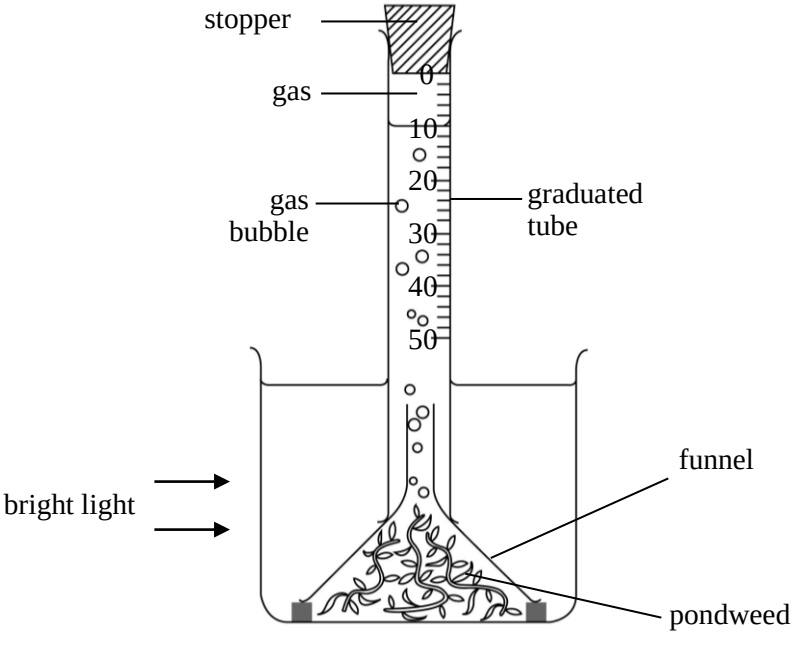
(12 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Presenting information	3
(c)	Analysing data	4
(d)(i)	Applying concepts	2
(d)(ii)	Applying concepts	2

|!|BCELB0210000019|!| ★ SI

The following diagram shows an experimental set-up, which included a graduated tube (graduated in cm^3) that was completely filled with water at the start of the experiment. After exposing the set-up to bright light for five hours, the water level in the tube dropped to the level shown in the diagram.



(a) Name the gas collected at the top of the tube and describe a test for it. (2 marks)

(b) Calculate the rate of gas release in the pondweed. Show your working. (2 marks)

(c) Suggest ***two*** ways to increase the rate of gas release in the pondweed. (2 marks)

(d) Predict the change in the volume of gas collected if a water snail is introduced into the funnel that holds the pondweed. Explain your answer. (3 marks)

##

(a) Oxygen (1)
Expose a glowing splint to the gas. The splint relights if the gas is oxygen. (1)

(b) Rate of gas evolution = 10 cm^3/ 5 hours (1)
= 2 cm^3/ hour (1)

(c) Any ***two*** of the following: (2)
- Increase the water temperature
- Increase the carbon dioxide concentration in the water by adding sodium hydrogencarbonate solution

(Or other reasonable answers)

(d) The volume of gas collected will decrease, (1)
because some of the oxygen produced by the pondweed will be consumed by the water snail (1)
to carry out respiration. (1)

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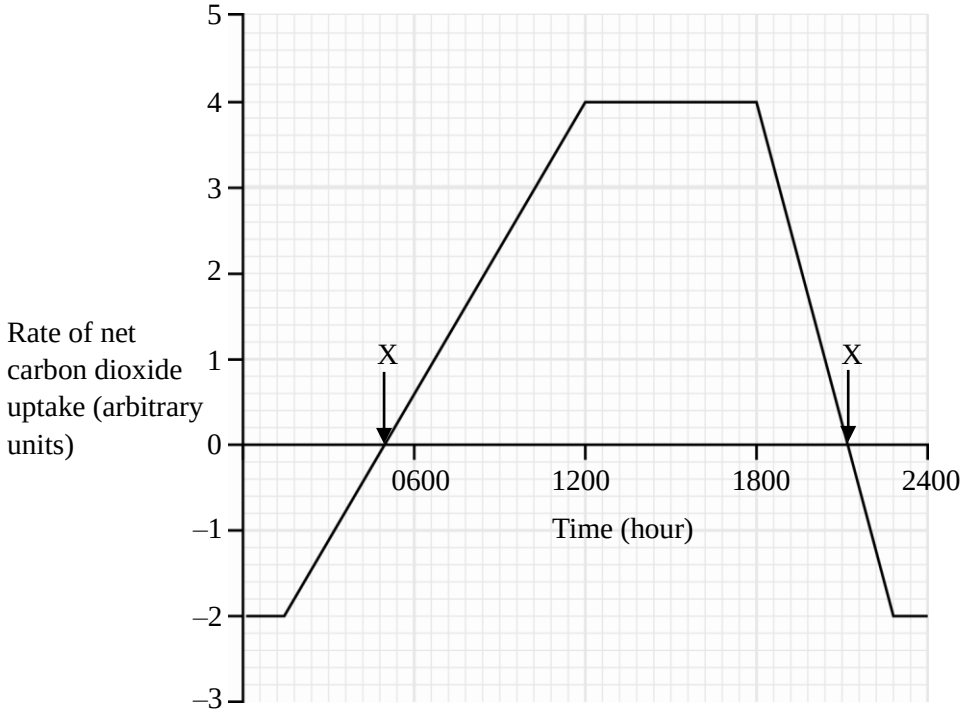
(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
	Practical skills	1
(b)	Mathematical skills	2
(c)	Practical skills	2
(d)	Applying concepts	3

|!|BCELA0210000020|!| ★★ **SI**

The following graph shows the changes in the rate of net carbon dioxide uptake by a crop in a greenhouse over a 24-hour period. The greenhouse receives sunlight and does not provide any artificial lighting.



(a) What causes the rate of net carbon dioxide uptake to become zero at time X?

(2 marks)

(b) Farmers can adjust the temperature in the greenhouse to increase crop yield.

(i) Explain why it would be economically inefficient to artificially increase the temperature throughout the entire 24-hour period. (3 marks)

(ii) Explain why it would be potentially beneficial to increase the temperature between noon and 6 p.m. (3 marks)

(c) Besides adjusting the temperature, suggest ***two*** ways to increase crop growth. (2 marks)

##

(a) When the rate of photosynthesis is equal to the rate of respiration, (1)
the rate of carbon dioxide uptake for photosynthesis also becomes equal to the rate of carbon dioxide release from respiration. (1)

(b) (i) During the early morning and evening periods, light is the main limiting factor for photosynthesis. (1)
Yet increased temperatures may also increase the rate of respiration. (1)
Thus raising the temperature in the aforementioned periods might cause more stored food (carbohydrates) to be used than will be produced. (1)

(ii) Between noon and 6 p.m., light is not the limiting factor of photosynthesis while temperature may limit the rate of photosynthesis. (1)
Raising the temperature could thus increase the activity of photosynthetic enzymes. (1)

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More food would therefore be produced and the growth rate of the crop would increase. (1)

(c) Any ***two*** of the following:

- Increase the concentration of carbon dioxide in the air (1)
- Increase soil fertility (1)

(Or other reasonable answers)

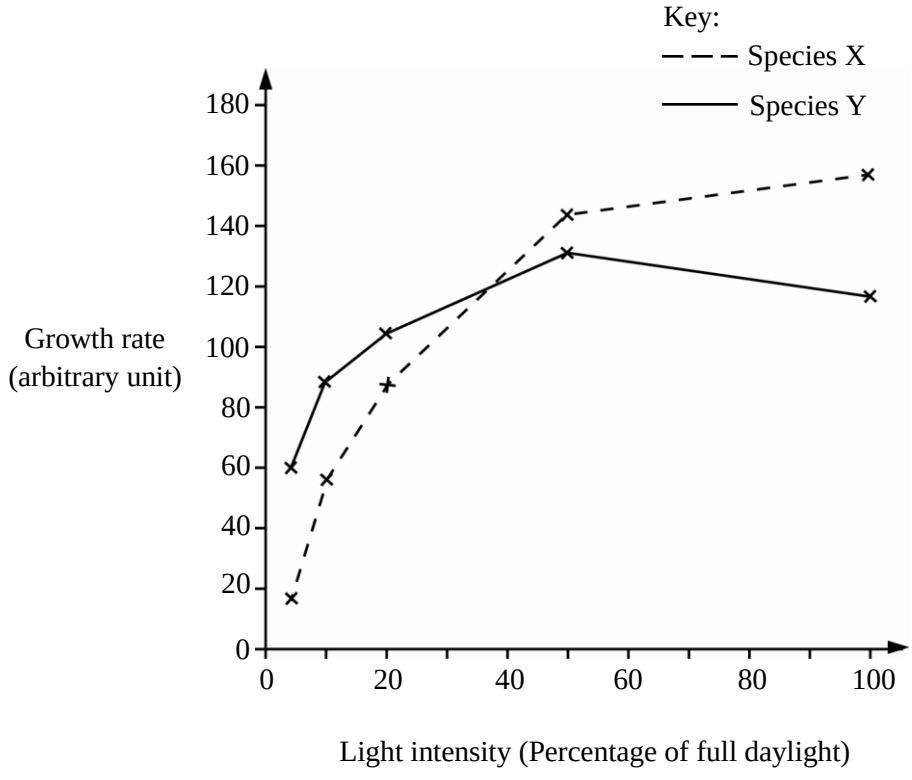
(10 marks)

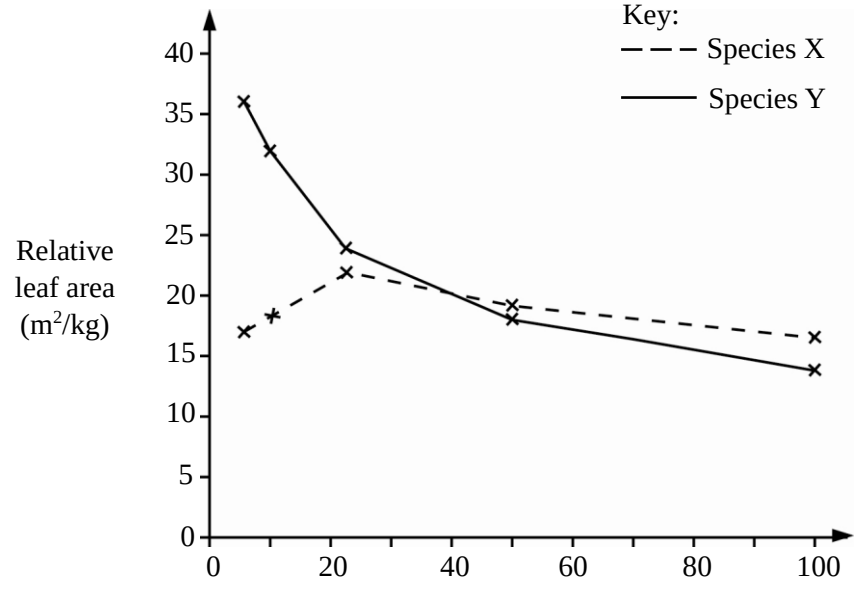
##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)(i)	Analysing data	1
	Applying concepts	2
(b)(ii)	Analysing data	1
	Applying concepts	2
(c)	Applying concepts	2

|!|BCELA0210000021|!| ★★

Zoe carried out an investigation to study the effect of light intensity on the growth of two plant species, X and Y. She measured the total growth rate and the relative leaf surface area of the two species under different light intensities. The following graphs show the results.





(a) According to the graphs above, state and explain which plant species is better adapted to woodland floors, where light intensity is low. (3 marks)

(b) Both of these plant species have a low growth rate at low light intensities. It may be due to a number of factors.

(i) Describe how low light intensities affect the gas exchange required for photosynthesis in plants. (2 marks)

(ii) Describe how low light intensities affect the main stages of photosynthesis. (4 marks)

##

(a) Species Y (1)
Its growth rate is greater at low light intensities. (1)
It has a larger relative leaf area and can thus absorb more light. (1)

(b) (i) At low light intensities, the stomata close. (1)
Less carbon dioxide is therefore taken up by the leaves. (1)

(ii) At low light intensities, less light is absorbed by chlorophylls. (1)
Thus less ATP / NADPH is formed in photochemical reactions. (1)
The reactions in the Calvin cycle are consequently inhibited (1)
and less carbohydrate is produced. (1)

(9 marks)

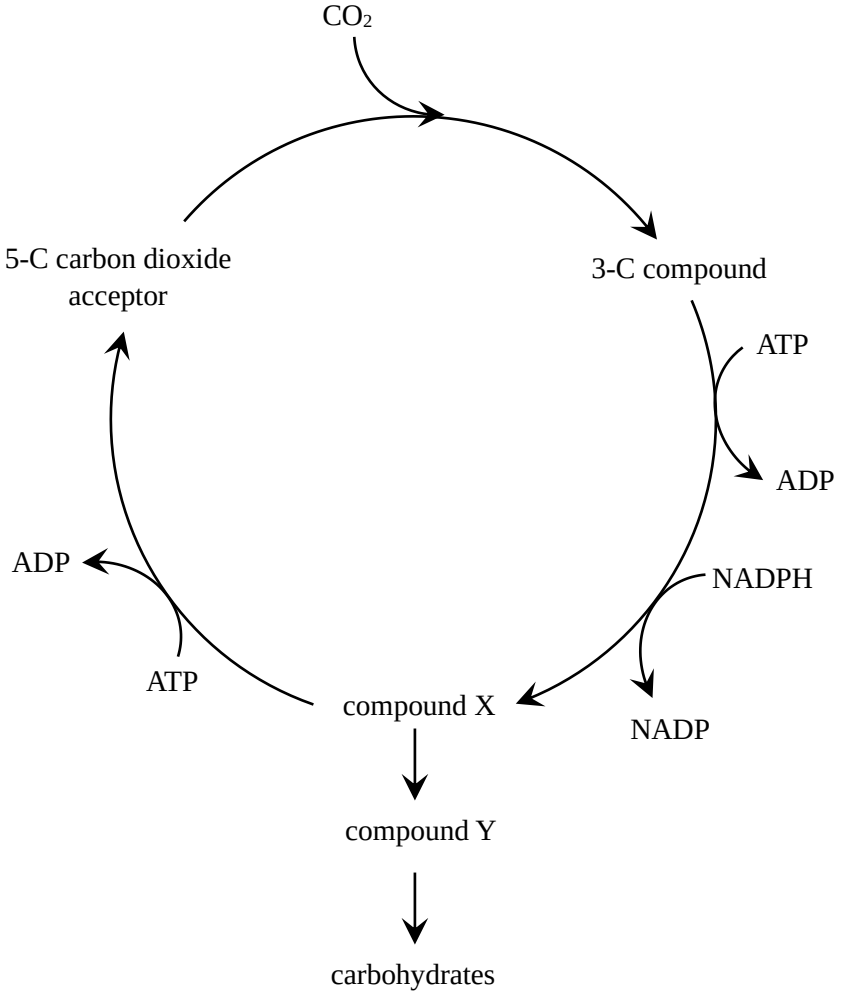
##

	Skills assessed	Marks
(a)	Analysing data	3
(b)(i)	Understanding basic concepts	2
(b)(ii)	Applying concepts	4

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|!|BCELB0210000022|!| ★

The following flowchart shows the Calvin cycle.



(a) State and explain ***two*** factors that affect the rate of the formation of the 3-C compound. (4 marks)

(b) Describe the roles of ATP and NADPH in converting the 3-C compound to compound X. (2 marks)

(c) Explain how the Calvin cycle will be affected if all of the compound X available is used to form compound Y. (3 marks)

(d) Suggest ***two*** functions of carbohydrates in plants. (2 marks)

##

(a) Any ***two*** of the following factors: (2)
- Light intensity
- Carbon dioxide concentration
- Temperature

Any ***two*** of the corresponding explanations: (2)
- Light is involved in photochemical reactions that produce ATP, which is essential for the regeneration of the 5-C carbon dioxide acceptor that is used to form the 3-C compound.
- Carbon dioxide is a raw material in the synthesis of the 3-C compound.
- Temperature will affect the reaction rate of the enzyme involved in the conversion of the 5-C carbon dioxide acceptor to the 3-C compound.

(b) ATP provides energy for the conversion of the 3-C compound to compound X. (1)
NADPH reduces the 3-C compound to compound X. (1)

(c) The Calvin cycle will stop (1)
because the regeneration of the 5-C carbon dioxide acceptor from compound X can no longer occur. (1)
The 5-C carbon dioxide acceptor is necessary to fix carbon dioxide and thus allow the continuation of the cycle. (1)

(d) Any ***two*** of the following: (2)
- Starch acts as a food reserve.
- Cellulose is used to make cell walls in new cells.

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- Glucose is used to release energy in respiration.

(Or other reasonable answers)

(11 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)	Understanding basic concepts	2
(c)	Applying concepts	3
(d)	Understanding basic concepts	2

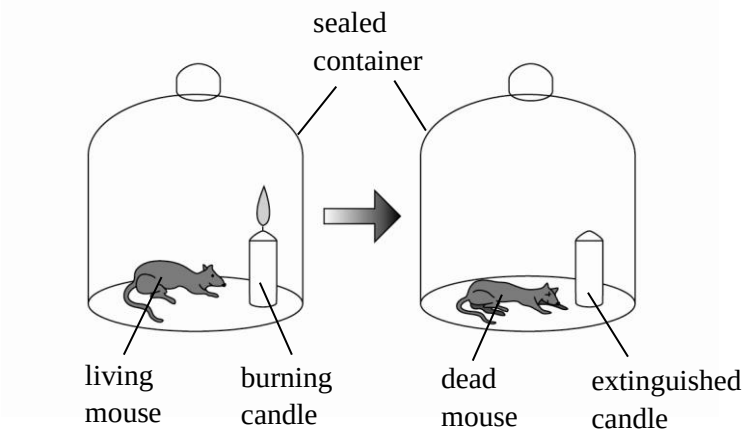
|!|BCELA0210000023|!| ★★ **SI** **NOS**

Joseph Priestley (1733–1804) was an English scientist. When he was young, he became curious how a spider kept in a sealed container would die. As an adult, he would investigate by carrying out the following experiments.

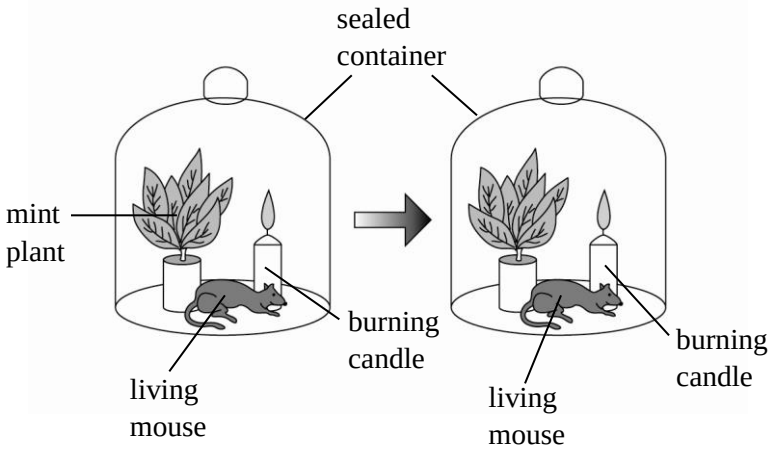
In Experiment 1, he put a mouse in a sealed container with a burning candle. The mouse would die after a period of time.

In Experiment 2, he put both a mint plant and a mouse in the sealed container. Priestley expected the organisms to die, but to his surprise they survived. Priestley would later discover oxygen gas.

Experiment 1



Experiment 2



(a) State the hypothesis proposed by Joseph Priestley before conducting Experiment 2. (1 mark)

(b) Based on your knowledge, explain the results

(i) in Experiment 1. (2 marks)

(ii) in Experiment 2. (2 marks)

(c) In the 1770s, the Dutch scientist Jan Ingenhousz became interested in the gas exchange in plants after reading about the work of Joseph Priestley. In 1779, Ingenhousz discovered that aquatic plants would give off bubbles from their green parts in the presence of light. He also found that the release of bubbles would eventually stop if the plants were in the shade. He identified the gas in the bubbles as oxygen.

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(i) Elaborate on how the work of Jan Ingenhousz exemplified the two aspects of the nature of science listed in the table below. (2 marks)

Nature of Science	***Elaboration***
Interpretation of observations is guided by our prior understanding of other theories and concepts.	
Science is a process of ongoing inquiries.	

(ii) Design an experimental set-up to replicate Ingenhousz's discovery mentioned above. (4 marks)

##

(a) The mint plant and the mouse will die in the sealed container with a burning candle. (1)

(b) (i) The burning candle consumed the oxygen inside the container. (1)
The mouse died due to suffocation / the shortage of oxygen. (1)

(ii) The mint plant carried out photosynthesis and released oxygen, (1)
which replenished the oxygen consumed in the burning of the candle and the respiration of the living mouse. (1)

(c) (i) 1 mark for each point (2)

Nature of Science	***Elaboration***
Interpretation of observations is guided by our prior understanding of other theories and concepts.	Jan Ingenhousz used the results from the experiments of Joseph Priestley to discover that photosynthesis requires light and the green parts of the plant.

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Science is a process of ongoing inquiries.	Joseph Priestley's work only provided some understanding about the release of oxygen in plants. Jan Ingenhousz's results helped further develop this scientific understanding.

(ii) Place a green aquatic plant in a transparent container of water. (1)
Expose the container to bright sunlight. (1)
Observe whether gas bubbles form around the leaves. (1)
Test the gas trapped in the container with a glowing splint to determine if the gas is oxygen. (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2
(c)(i)	Critical thinking	2
(c)(ii)	Practical skills	4

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